Autonomous Systems Safety: Ethical Decision-Making in Self-Driving Vehicles

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***Abstract*—**Autonomous systems are increasingly being deployed in environments where decision-making can have direct consequences on human life. This paper examines how different ethical frameworks influence decision-making in self-driving vehicles, particularly in high-risk scenarios involving unavoidable harm. Using a simulated autonomous driving environment, three distinct moral decision models: humanist, protectionist, and profit-based, are evaluated across controlled scenarios.

The analysis focuses on how these frameworks behave not only in isolation but also in multi-agent contexts, where multiple autonomous systems interact. The findings highlight the limitations of single-agent ethical reasoning, the instability of certain decision rules in shared environments, and the misalignment between legal compliance and ethical outcomes. The study further explores implications for policy, system design, and real-world deployment of autonomous systems.

1 INTRODUCTION

The deployment of autonomous systems in real-world environments introduces a fundamental challenge: how should machines make decisions in scenarios involving moral trade-offs? In domains such as self-driving vehicles, these decisions can involve life-and-death consequences, requiring systems to evaluate competing outcomes under uncertainty.

Traditional approaches to autonomous decision-making focus on optimization, safety, and efficiency. However, in edge-case scenarios such as unavoidable collisions, these systems must also implicitly encode ethical judgments. This raises

questions about whether ethical reasoning can be formalized computationally, and if so, what principles should guide these decisions.

This paper examines ethical decision-making in autonomous vehicles using simulated scenarios designed to replicate high-risk situations. The goal is to analyze how different moral frameworks behave under controlled conditions and to identify their limitations when applied to real-world environments.

2 METHODOLOGY

This study uses a simulation-based approach to evaluate ethical decision-making in autonomous vehicles. The analysis is based on a predefined simulation environment in which a self-driving car encounters various moral dilemmas.

2.1 Ethical Frameworks

Three decision-making models are embedded within the system:

- **Humanist Model**: Prioritizes minimizing total harm across all affected individuals
- **Protectionist Model**: Prioritizes the safety of the vehicle's occupants
- **Profit-Based Model**: Prioritizes minimizing financial cost or liability

Each model processes the same scenario but produces different outcomes based on its underlying objective function.

2.2 Scenario Selection

The analysis focuses on the **school bus scenario**, which involves a trade-off between protecting the vehicle's occupant and minimizing harm to a larger group of external individuals.

This scenario was selected because it presents:

- asymmetric risk (one vs many individuals)
- vulnerability differences (children vs adult driver)
- clear ethical conflict between competing frameworks

2.3 Analytical Approach

The evaluation is conducted across four dimensions:

1. Framework alignment with real-world industry positions
2. Behaviour under multi-agent interaction (system interacting with itself)
3. Impact of communication between autonomous systems
4. Relationship between legal rules and ethical outcomes

Rather than evaluating correctness, the analysis focuses on:

- consistency
- stability
- scalability of ethical decision-making

3 RESULTS AND ANALYSIS

3.1 3A: Mercedes DRIVERLESS CARS

Scenario Chosen: School Bus

The 2016 claim by Mercedes that its autonomous vehicles would always prioritize the safety of passengers aligns most closely with the Protectionist algorithm in the simulation. The protectionist model explicitly prioritizes the vehicle's occupant over all external parties, regardless of the number or vulnerability of others involved. In the school bus scenario, this meant that the autonomous vehicle selected the path that maximized the survival probability of the 24-year-old Male driver, even when that choice increased risk to the sixteen children on the bus and with two adults. The ethical framework here is not harm minimization, but user preservation.

If the school bus were replaced with another autonomous vehicle operating under the same protectionist algorithm, the outcome becomes more complex. Both vehicles would prioritize their own occupants, and neither would willingly sacrifice its passenger for the other. This creates a form of ethical deadlock. Without a mechanism for shared moral reasoning, both systems may take aggressive

protective actions that increase overall harm. In effect, the passenger-first rule becomes unstable when interacting with itself.

If the vehicles were capable of communicating, the outcome might improve slightly, but only if communication allows for coordination beyond self-interest. If both vehicles still adhere strictly to a protectionist rule, communication would not meaningfully change the decision because neither vehicle would concede passenger safety. However, if communication allowed vehicles to share injury probability data and coordinate to minimize overall harm, then a more balanced outcome could emerge.

Allowing vehicles to communicate raises serious data privacy concerns. In order to coordinate effectively, vehicles would need to exchange sensitive information such as passenger age, identity, insurance status, or risk profiles. This introduces risks of misuse, discrimination, or unauthorized data access. If vehicles begin assigning value based on insurance coverage or social identity, ethical decision-making becomes intertwined with economic and personal data, which raises significant concerns about surveillance and inequality.

3.2 3B: TESLA TROLLY TYPE CRASH DILEMMA

The Tesla approach described in the 2017 article is closest to the Protectionist algorithm. By refusing to retake control of the wheel even if the driver is deliberately steering toward a group of people the system prioritizes the autonomy and agency of the driver over external harm reduction. In practice, this means the vehicle does not override the interests or intentions of its user.

Unlike the Humanist algorithm, which would intervene to minimize total casualties, Tesla's implied approach avoids redistributing harm if doing so would require overriding the driver's control. It also does not resemble the Profit algorithm, which makes decisions based on financial cost or liability calculations. Instead, the Tesla stance aligns most closely with the Protectionist principle because it centers the vehicle's occupant and refrains from sacrificing their control or safety for bystanders.

3.3 3C: MORAL ADAPTATION ACROSS COUNTRIES

If a self-driving car operates in multiple countries with different driving norms and legal systems, the ethical framework guiding its decisions should remain consistent rather than adapt to local moral preferences or allow user-controlled switching. While traffic laws and operational parameters must adapt to local regulations, the underlying moral judgment algorithm should not vary by country or individual user selection.

Allowing moral adaptation based on geography risks turning ethical principles into culturally negotiable variables. A human life should not be valued differently depending on national context. Similarly, allowing drivers to switch between moral settings would effectively commercialize ethical decision-making, enabling individuals to choose self-serving frameworks such as passenger-first protection over broader harm minimization.

A consistent moral framework such as the Humanist principle that seeks to minimize overall harm provides greater predictability and fairness. While implementation details (such as driving customs or legal compliance) should adjust to local norms, the core ethical logic should remain stable to avoid inequality, manipulation, or morally inconsistent outcomes across jurisdictions.

3.4 3D: rULE FOLLOWING STARTUP

If a self-driving startup decided its vehicles would always strictly follow traffic laws, the outcome in the school bus scenario would depend on which moral principle is being applied.

In the simulation, the Humanist algorithm did not violate any traffic rules. It remained in its lane and chose a frontal collision that resulted in mild injuries. In this case, ethical harm minimization aligned with legal compliance.

By contrast, the Protectionist algorithm violated traffic norms by swerving into another lane to protect the 24-year-old driver, even though this increased the risk to the bus and its occupants. Similarly, the Profit algorithm left the roadway entirely, prioritizing cost minimization over passenger safety. This maneuver required illegal movement and resulted in harm to the driver in order to reduce financial liability.

If all traffic rules were strictly followed in this scenario, the outcome would resemble the Humanist decision, since it remained within its lane and did not require illegal maneuvers. However, the broader lesson remains that moral reasoning and rule compliance do not always align. While in this particular case the humanitarian choice was also legally compliant, other emergency scenarios may require rule violations to reduce harm. Therefore, strict rule-following does not guarantee the most ethical outcome in every case.

4 CONCLUSION

The analysis demonstrates that ethical decision-making in autonomous systems cannot be treated as a purely isolated or rule-based problem. While individual frameworks such as humanist, protectionist, and profit-based models produce internally consistent decisions, their behavior becomes significantly more complex when deployed in shared, real-world environments.

In particular, the protectionist model highlights a critical limitation: decision rules that prioritize self-preservation may become unstable when interacting with similar systems, leading to conflict and increased overall harm. This suggests that evaluating autonomous systems solely at the individual level is insufficient, and that multi-agent interactions must be considered as a core component of safety analysis.

The findings also show that communication between systems does not inherently resolve ethical conflicts unless underlying decision frameworks allow for coordination beyond self-interest. Similarly, strict adherence to legal rules does not guarantee ethical outcomes in high-risk scenarios, indicating a gap between regulatory compliance and moral reasoning.

Finally, the question of whether ethical frameworks should adapt across countries reveals a broader challenge: allowing variability in moral logic risks inconsistency, inequality, and manipulation. A stable and predictable ethical foundation is therefore necessary, even as operational behaviors adapt to local contexts.

Overall, this study highlights the need to move beyond static ethical rules toward more robust, system-level approaches that account for interaction, coordination, and real-world complexity in autonomous environments.